COMPUTER ENGINEERING

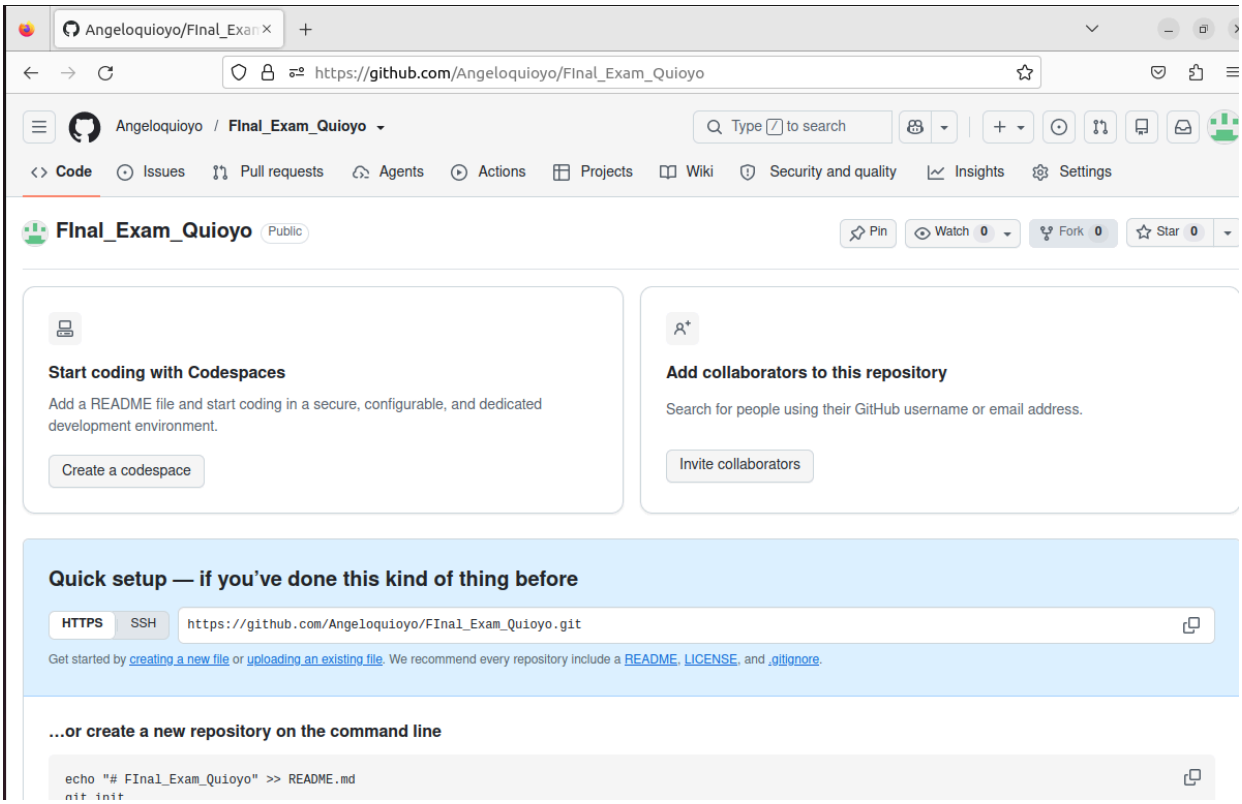
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COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/29/2026	**DATE SUBMITTED:**	10/29/2026
NAME:	Quioyo, Angelo M.	**INSTRUCTOR:**	Engr. Jimlord Quejado

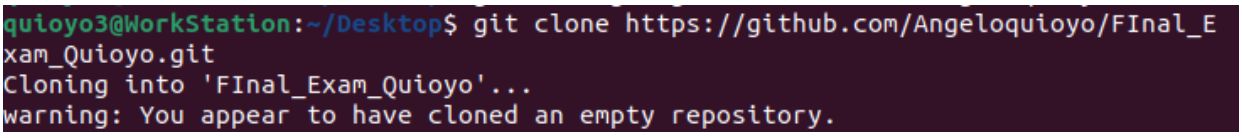
ACTIVITY NO. **Final Skills Exam**

1. PROCEDURE OUTPUT

1. Create a repository and label it as "Final_Exam_Surname"



2. Clone your new repository in your VM



```
quioyo3@WorkStation:~/Desktop$ cd FInal_Exam_Quioyo
quioyo3@WorkStation:~/Desktop/FInal_Exam_Quioyo$
```

3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.

```
quioyo3@WorkStation:~/Desktop/FInal_Exam_Quioyo$ cat inventory.ini
[debian_servers]
ubuntu-node ansible_host=192.168.56.110 ansible_user=ubuntu

[centos_servers]
centos-node ansible_host=10.0.2.15 ansible_user=centos

[all_servers:children]
debian_servers
centos_servers
```

3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers

3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)

```
quioyo3@WorkStation:~/Desktop/FInal_Exam_Quioyo$ nano site.yaml
quioyo3@WorkStation:~/Desktop/FInal_Exam_Quioyo$ cat site.yaml
---
- name: Configure Enterprise Service, Monitoring, and MOTD
  hosts: all
  become: true
  vars_files:
    - config.yaml

  tasks:
    - name: Install Nginx (Debian/Ubuntu)
      ansible.builtin.apt:
        name: "{{ enterprise_service }}"
        state: present
        update_cache: yes
      when: ansible_os_family == "Debian"

    - name: Install Nginx (CentOS/RHEL)
      ansible.builtin.dnf:
        name: "{{ enterprise_service }}"
        state: present
      when: ansible_os_family == "RedHat"

    - name: Ensure Nginx service is running and enabled
      ansible.builtin.service:
        name: "{{ enterprise_service }}"
        state: started
        enabled: yes

    - name: Install Node Exporter (Debian/Ubuntu)
      ansible.builtin.apt:
        name: prometheus-node-exporter
        state: present
      when: ansible_os_family == "Debian"

    - name: Install Node Exporter (CentOS/RHEL)
      ansible.builtin.dnf:
        name: golang-github-prometheus-node-exporter
        state: present
      when: ansible_os_family == "RedHat"
```

```
    - name: Ensure Nginx service is running and enabled
      ansible.builtin.service:
        name: "{{ enterprise_service }}"
        state: started
        enabled: yes

    - name: Install Node Exporter (Debian/Ubuntu)
      ansible.builtin.apt:
        name: prometheus-node-exporter
        state: present
      when: ansible_os_family == "Debian"

    - name: Install Node Exporter (CentOS/RHEL)
      ansible.builtin.dnf:
        name: golang-github-prometheus-node-exporter
        state: present
      when: ansible_os_family == "RedHat"

    - name: Ensure Monitoring Tool service is running and enabled
      ansible.builtin.service:
        name: "{{ 'prometheus-node-exporter' if ansible_os_family == 'Debian' el
se 'node_exporter' }}"
        state: started
        enabled: yes

    - name: Update MOTD file
      ansible.builtin.template:
        src: templates/motd.j2
        dest: /etc/motd
        owner: root
        group: root
        mode: '0644'
quioyo3@WorkStation:~/Desktop/FInal_Exam_Quioyo$
```

4.4 Change Motd as "Ansible Managed by <username>"

```
quioyo3@WorkStation:~/Desktop/FInal_Exam_Quioyo$ cat templates/motd.j2
Ansible Managed by {{ Angelo }}
```

```
PLAY RECAP *********************************************************************
***
localhost                  : ok=3    changed=1    unreachable=0    failed=1
  skipped=1    rescued=0    ignored=0

quioyo3@WorkStation:~/Desktop/FInal_Exam_Quioyo$
```

4. Push and commit your files in GitHub

```
quioyo3@WorkStation:~/Desktop/FInal_Exam_Quioyo$ git push origin main
Username for 'https://github.com': Angeloquioyo
Password for 'https://Angeloquioyo@github.com':
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 4 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 1.21 KiB | 1.21 MiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Angeloquioyo/FInal_Exam_Quioyo.git
 * [new branch]      main -> main
quioyo3@WorkStation:~/Desktop/FInal_Exam_Quioyo$
```

5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)

5. For your final exam to be counted, please paste your repository link as an answer in this exam.

https://github.com/Angeloquioyo/FInal_Exam_Quioyo

Note: Extra points if you will implement the said services via containerization.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

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Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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